

Design of Python-based Schedule Management System (Grade: A)

Background of the problem

In daily life and work, we often need to arrange and manage our schedules to ensure the completion of tasks and the rational use of time. To improve the efficiency and convenience of schedule management, a simple schedule management system based on Python is designed, which can realize functions such as adding, viewing, modifying and deleting schedules.

Task Description

Design a simple and visualized schedule management system with the following functions:

- User interface: Design an intuitive and user-friendly interface that includes a schedule list, calendar selection, and functional buttons such as add, view, modify, and delete.
- Schedule addition: Users can input the title, date, time and remarks of the schedule through the interface and save it to the system.
- Schedule Viewing: Users can select a date through the interface to view all the schedules for that day, which will be displayed on the interface.
- Schedule Modification: Users can select a specific schedule for modification and update the details of the schedule in the interface.
- Schedule Deletion: Users can select a specific schedule to delete, ensuring the safety and accuracy of the deletion operation.
- Error Handling: Provide friendly prompt messages for illegal operations or abnormal situations.

Hint

- The graphical user interface can be implemented using libraries such as PyQt and Tkinter.
- Considering the interface aesthetics and user experience, make the interface as simple and clear as possible.
 - Consider handling exceptional situations to ensure the stability of the program and user-friendliness.
 - The Python datetime library can be used to handle date and time information, ensuring the accuracy and reliability of the schedule.
 - A calendar selection function can be added to the interface to facilitate users' selection of dates.
 - Python's list or dictionary and other data structures can be used to organize and store schedule information, enabling the functions of adding, deleting, modifying and querying schedules.

document

Each group should submit the following materials:

- Each person should create a Word design document: clearly state each person's responsibilities and precisely indicate who completed which part. Some overlap is acceptable. Focus on your own contributions while ensuring the document's completeness. Follow the specified naming format for the file. The report should include flowcharts, diagrams, statistical charts, screenshots of tests, etc., for explanation. The report should be at least **20 pages** long, excluding appendices.
- One PowerPoint presentation: It should include charts, implementation methods, etc. The first page should clearly list the names and student numbers of the group members, sorted by contribution size. Each person's specific contribution should also be marked within the slides. The presentation should **be no less than 20** pages.
- The source code is placed in the src directory. If any third-party packages are used, they should be placed in the lib subdirectory under the src directory.
- README file: Describes the environment configuration, compilation, execution, testing, and results of the source code, etc.

Attention

Common reasons for **not passing**: self-proposed questions, *implementation in non-Python languages*, *plagiarism or being plagiarized*, failure to complete basic functions, lack of the required documents mentioned above, not doing the assignment or not submitting it...